CRYSS V3.0 — Concept & Roadmap (Public Disclosure)

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1. Motivation for CRYSS V3.0

CRYSS V2.0 enables rapid, mixture-centric thermodynamic modelling using parallel experimentation data. When combined with high-throughput platforms, users can quickly identify which systems exhibit genuine purification potential.

However, once promising systems are identified, deeper insight is required:

- How sensitive is the system to feedstock composition?
- How do solvent effects modulate Rmax?
- How stable is the purification window across solvent space?
- How do fitted PhyChem parameters relate to underlying molecular behaviour?
- Can we map “danger zones” where purification collapses?

CRYSS V3.0 addresses these questions by extending CRYSS into a **high-resolution, sensitivity-aware modelling framework**.

2. Overview of the CRYSS V3.0 Workflow

CRYSS V3.0 introduces a two-stage modelling pipeline:

Stage 1 — System Ranking (CRYSS V2.0 + Hardware)

Parallel experimentation generates mixture data across solvent and composition space. CRYSS V2.0 fits T, ΔH, and Mode parameters (incorporating deviations from ideality via lumped activity coefficients). This enables ranking of systems worth deeper exploration.

Stage 2 — High-Resolution Profiling (CRYSS V3.0)

Selected systems undergo multi-solvent slice scanning. The more slices collected, the more accurately CRYSS can map the underlying PhyChem landscape. This produces high-resolution fitted parameters with meaningful physical interpretation.

3. High-Resolution PhyChem Parameter Extraction

CRYSS V3.0 leverages the fact that:

- The fitted T, ΔH, and Mode parameters encode deviations from ideality.
- These deviations arise from underlying activity coefficients.
- Multi-solvent scanning increases the information density.
- The relative values of T, ΔH, Mode become physically interpretable when fitted across many slices.

Thus, CRYSS V3.0 treats the fitted parameters as **latent PhyChem descriptors** of the system.

These descriptors can be used to:

- map solvent sensitivity
- identify solvent-dependent eutectic behaviour
- quantify robustness of purification windows
- detect composition-dependent failure modes

This transforms CRYSS from a "single-slice solver" into a **multi-slice PhyChem high resolution process mapper**.

4. Sensitivity Mapping and Rmax Response Surfaces

CRYSS V3.0 introduces a new modelling layer:

Rmax Sensitivity Simulation

Using the high-resolution fitted parameters, CRYSS V3.0 simulates Rmax across:

- initial composition space
- solvent space
- multi-component mixtures
- impurity loadings
- feedstock variability

This produces **response surfaces** showing:

- regions of high purification potential
- regions where purification collapses
- "tricky" feedstock compositions
- solvent-dependent vulnerabilities
- impurity-driven shifts in Rmax

These surfaces allow users to:

- predict failure modes before experimentation
- design robust purification strategies
- avoid composition regions with deleterious effects
- understand how solvent choice modulates thermodynamic behaviour

5. Correlation Analysis and PCA Mapping

CRYSS V3.0 introduces a correlation-mapping module to unpick relationships between:

- fitted PhyChem parameters
- solvent descriptors
- Rmax sensitivity
- composition robustness
- impurity tolerance

Two approaches are envisioned:

A. Response Surface Correlation Maps

Visualising how Rmax sensitivity correlates with PhyChem parameters.

B. PCA / Dimensionality Reduction Maps

Projecting systems into a low-dimensional space to reveal:

- clusters of robust systems
- clusters of sensitive systems
- solvent-driven shifts
- PhyChem-driven vulnerabilities

This provides a **global view** of how systems behave across solvent and composition space.

6. Future Direction (CRYSS V4.0)

This section is intentionally conceptual and non-binding.

CRYSS V4.0 may explore separating the activity coefficient contributions explicitly, enabling:

- direct mapping of solvent effects onto solvent descriptor PCA space
- hybrid mechanistic + descriptor-driven modelling
- solvent-aware prediction of eutectic behaviour
- cross-solvent generalization
- solvent selection algorithms
- descriptor-driven optimization

This would unify:

- the CRYSS DoE solvent PCA module
- the CRYSS core thermodynamic engine
- the CRYSS multi-slice PhyChem mapper

into a single **descriptor-aware purification prediction platform**.

7. Integration with High-Throughput Platforms

CRYSS V3.0 is designed to integrate seamlessly with parallel experimentation systems, enabling:

- automated ingestion of multi-solvent slice data
- real-time fitting of PhyChem parameters
- automated sensitivity mapping
- closed-loop optimisation
- platform-specific visualisation modules
- API-based communication between hardware and CRYSS

This positions CRYSS as the **thermodynamic intelligence layer** for high-throughput purification workflows.

8. Licensing and Open-Source Position

CRYSS V3.0 will remain MIT-licensed, ensuring:

- open access
- collaborative development
- freedom to operate
- protection via public prior art
- compatibility with academic and industrial workflows

This document establishes the conceptual foundation for CRYSS V3.0 as public prior art.

9. Summary

CRYSS V3.0 extends CRYSS from a mixture-centric solver into a **high-resolution, sensitivity-aware, descriptor-aware thermodynamic engine** capable of:

- ranking systems
- profiling PhyChem behaviour
- mapping sensitivity
- predicting failure modes
- correlating solvent effects
- integrating with high-throughput platforms

This roadmap positions CRYSS as a next-generation purification modelling platform with unique capabilities in the chemical development landscape.